

Stock Market Reaction on Green-Bond Issue: Evidence from Indian Green-Bond Issuers

Vision

27(2) 264–272, 2023

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DOI: 10.1177/09722629211022523

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Abstract

Purpose: A green bond is a financial instrument issued by governments, financial institutions and corporations to fund green projects, such as those involving renewable energy, green buildings, low carbon transport, etc. This study analyses the effect of green-bond issue announcement on the issuer's stock price movement. It shows the reaction of the stock price after the issue of green bonds.

Methodology: This study is based on secondary data. Green-bond issue dates have been collected from newspaper articles from different online sources, such as *Business Standard*, *The Economic Times*, Moneycontrol, etc. The closing prices of stocks have been taken from the NSE (National Stock Exchange of India Limited) website. An event window of 21 days has been fixed for the study, including the 10 days before and after the issue date. Data analysis is carried out through the event study method using the R software. Calculation of abnormal returns is done using three models: mean-adjusted returns model, market-adjusted returns model and risk-adjusted returns model.

Findings: The results show that the issue of green bonds has a significant positive effect on the stock price. Returns increase after the green-bond issue announcement. Although the announcement day shows a negative return for all the samples taken for the study, the 10-day cumulative abnormal return (CAR) is positive. Thus, green-bond issues lead to positive sentiments among investors.

Research implications: This research article will help the government issue more green bonds so that the proceeds can be utilized for green projects. The government should motivate corporations and financial institutions to issue more green bonds to help the economy grow. In India, very few organizations have issued a green bond. It will be beneficial if these players issue green bonds, as it will increase the firms' value and boost returns to the investors.

Originality/value: The effect of green-bond issue on stock returns has been analysed in some studies in developed countries. This is the first study to examine the impact of green-bond issue on stock returns in the Indian context, to the best of our knowledge.

Key Words

Sustainable Development, Green Bonds, Stock Returns, Event Study Method

Executive Summary

A green bond is a financial instrument that is issued by governments, financial institutions and corporations to fund green projects, such as those involving renewable energy, green buildings, low carbon transport, etc. This study analysed the effect of green-bond issue announcement on the movement of the stock price of the issuer. It shows the reaction of the stock price after the issue of green bonds. The study is based on secondary data. Green-bond issue dates were collected from newspaper articles from

different online sources, such as *Business Standard*, *The Economic Times*, Moneycontrol, etc. The closing prices of stocks have been taken from the NSE (National Stock Exchange of India Limited) website. An event window of 21 days was fixed for the study, that is, including the 10 days before and after the issue date. Data analysis was carried out using the event study method with the help of R software. Calculation of abnormal returns was done using three models: mean-adjusted returns model, market-adjusted returns model and risk-adjusted returns model. The results show that the issue of green bonds has a

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significant positive effect on the stock price. Returns increased after the announcement of green-bond issue; although the announcement day showed a negative return for all the samples taken for the study, the 10-day cumulative abnormal return (CAR) was found positive. Thus, green-bond issues lead to positive sentiments among investors. This research article will help the government issue more green bonds so that proceeds can be utilized for green projects. Further, the government should motivate corporations and financial institutions to issue more green bonds for the growth of the economy. In India, very few organizations have issued a green bond. It will be beneficial if these players issue green bonds, as it will increase the value of the firms and boost returns to the investors. The effect of green-bond issue on stock returns has been analysed in some studies in developed countries. To the best of our knowledge, this is the first study to analyse the effect of green-bond issue on stock returns in the Indian context.

Introduction

Sustainable development and environmental protection are significant concerns of this time. However, different countries of the world are coming together and trying to formulate policy frameworks for reducing the impact of the greenhouse effect at the global level. Implementing and achieving the goals and targets of these frameworks is the most critical task (Opinion, 2020). The issue of green bonds to finance green projects is one of the significant reform steps towards protecting the environment and sustainable growth. These bonds help minimize risks, enhance a firm's image and improve the environment's quality. For example, for the purchase of a hybrid electric car, financing is provided by Toyota using green bonds (Climate Bonds Initiative, 2019). A green bond is a financial instrument like other conventional bonds, but the difference is that the utilization of funds raised from such bonds will be for the sole purpose of financing green projects (Kaminker, 2015). The proceeds from green bonds can finance projects involving renewable energy, sustainable energy, clean transport, sustainable water management, climate change adoption, green buildings, sustainable waste management, sustainable land use and biodiversity conservation (SEBI, 2017).

International Capital Markets Association (ICMA) has provided the list of projects that will come under the purview of green projects. India (Securities and Exchange Board of India [SEBI]) has also formulated guidelines for the issue of green bonds, eligible green projects and the bonds' utilization, with further scope to include other projects not specified in the list. The European Investment Bank (ECB) issued the first Green bond in 2007 worth Euro 600 million (Cochu et. al., 2016), followed by the World Bank in 2008 (De Mariz & Deschryver, 2020). In India, a green bond was first issued by Yes Bank in 2015, and in subsequent years, other financial institutions, such

Table 1. Issue and Oversubscription of Green Bonds in India.

Company (issue)	Oversubscription (times)
IREDA (2016)	5.10
Rural Electrification Corporation (REC)	3.90
EXIM Bank	3.00
IRFC	3.00
Axis Bank	2.20
Yes Bank	2.00
AZURE Power	2.00
IREDA (2017)	1.74
Greenko Group	1.50
NTPC	1.45

Source: CARE Ratings.

as Exim Bank, IDBI Bank, Axis Bank, State Bank of India (SBI), Rural Electrification Corporation (REC), Power Finance Corporation (PFC) and Indian Railway Finance Corporation, also issued green bonds. Indian financial institutions are the pioneers among all the issuers of green bonds in India. Yes Bank has raised funds through issuing green bonds two times. The Indian green-bond issues are oversubscribed, which shows the investors' concerns for the environment (Agarwal & Singh, 2018). In India, green bonds have attracted the attention of investors. Table 1 clearly shows their inclination towards investment in such bonds. On average, all the issues have been oversubscribed by more than 2.5 times.

Figure 1 shows that, as of 30 June 2019, renewable energy, energy efficiency and clean transportation projects received the most considerable portion, approximately 66%, out of the total green-bond commitments in the world. The use of proceeds from green bonds allocated for agriculture, land use, forests and ecological resources was 17%, while 8% was used for water and wastewater and resilient infrastructure. The least portion was allocated for solid waste management: 1%.

Green bonds provide many benefits to their stakeholders. First, the lenders and banks can maintain their assets–liability balance through issuing such bonds. Second, states can attract long-term foreign investors to increase their foreign reserves, improve energy security and reduce climate change's negative impacts. Third, green-bonds investors' image is enhanced through their orientation towards the environment and climate.

Table 2 shows how the proceeds from green bonds have been utilized in different sectors in India. From the table, we find that banking and financial institutions, such as Axis Bank, SBI, IDBI Bank, Exim Bank, PFC, Infrastructure Leasing & Financial Services, have issued most of the green bonds.

We observe that green bonds are becoming a popular source of finance in India. From Table 2, it can also be observed that most green-bond issuers are banks and financial institutions. There is further scope for the issue of

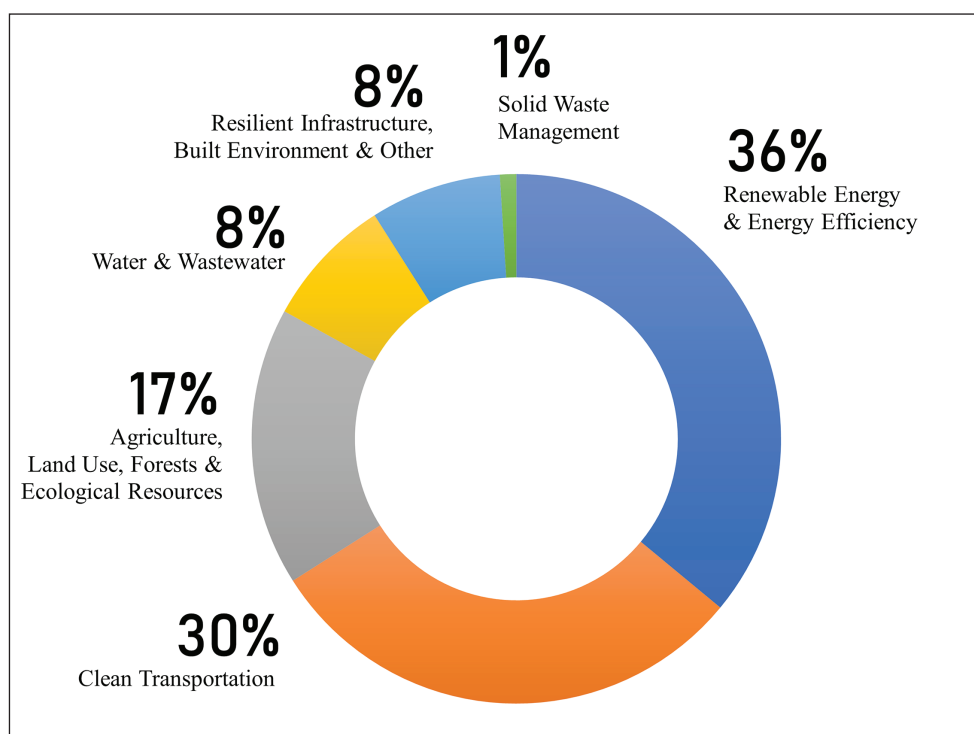


Figure 1. Green Bond-eligible Projects Portfolio.

Source: World Bank Green Bond Impact Report 2019.

Table 2. Utilization of Green Bonds in Different Sectors.

	Renewable Energy	Energy Efficiency	Low Carbon Transport	Low-carbon Buildings	Sustainable Water Management	Sustainable Waste Management	Energy Sector Technologies	Sustainable Land Use
Axis Bank	Yes		Yes	Yes				
NTPC	Yes							
REC	Yes	Yes			Yes	Yes		
SBI	Yes	Yes	Yes	Yes		Yes		
Greenko	Yes							
PFC	Yes	Yes	Yes			Yes	Yes	
IDBI Bank	Yes	Yes	Yes		Yes	Yes	Yes	Yes
CLP	Yes							
Exim Bank	Yes		Yes					
Hero Future Energies	Yes							
IRFC (Indian Railways)			Yes					
PNB				Yes				
L&T	Yes							
Azure Power	Yes							
Yes Bank	Yes							
ReNew Power	Yes							
IREDA								
IL&FS	Yes							
EESL		Yes						
Jain Irrigation					Yes			

Source: The Energy and Resource Institute Analysis.

green bonds by companies working in other sectors, such as pharmaceuticals, energy, automobiles, fast-moving consumer goods (FMCG), etc. This raises the question: What is the reason for the low number of green-bond issuers in India? This requires an analysis of how green bonds affect the performance of a company. This study investigates the effect of green-bond issues on the Indian companies issuing them, that is, how the stock returns of these companies get affected by their issuing of green bonds. The study's outcomes may guide companies working in different sectors in India to issue green bonds.

Banks and financial institutions whose stocks are listed on the stock exchange have been considered for the study. The event study method has been applied for data analysis, which shows the significant positive effect of green-bond issue on the stock returns during a 21-day event window (including the 10 days before and after the issue date). This study will help the corporations issue more green bonds, as these bonds are creating more wealth for the investors, while at the same time providing a better environment for future generations.

Literature Review

This article discusses corporate social responsibility (CSR) activities as a way for a business to demonstrate its commitment to society and the environment, in addition to its primary business. Raising capital through green bonds and reinvesting it solely in green projects is regarded as one of the issuer's social responsibilities. The article also focuses on how an investor can take the lead in environmental protection through financing and investing in these projects. Many studies show that the good performance of CSR activities enhances the stakeholders' engagement (Benabou & Tirole, 2010). A study by Lee and Faff (2009) shows that firms with higher CSR scores reduce idiosyncratic risks. Goss and Roberts (2011) found that firms that are less engaged in CSR activities pay more on borrowings—ranging from 7 to 18 basis points—than those highly engaged in such activities. Also, Dhaliwal et al. (2011) showed that voluntary disclosure of CSR activities has a significant role in reducing capital costs. Further, El Ghouli et al. (2011) found that it also reduces equity costs. Campbell and Taksler (2003) found that the corporate yield increases with an increase in volatility. Their study also revealed a positive relationship between the volatility of equity and the issuing company's borrowing cost. Greenwood and Vayanos (2014) and Collin-Dufresne et al. (2001) examined the factors affecting the bond yield and return, such as demand, supply and maturity structure. Bakshi and Preclaw (2015) found that investors are inclined towards green bonds when the volatility of the green bonds and similar conventional bonds are equal. Their study was conducted on corporate bond companies. This signifies that investors are expecting at least the same risk and return from green bonds too. Ehlers and Packer (2017) stated in their study that the spread of

green bonds is relatively lower than that of conventional bonds. This may be due to the higher demand for green bonds. The higher demand for such bonds shows investors' active participation and contribution to environment-friendly projects. Gianfrate and Peri (2019) explored the advantages of issuing a green bond. Using the propensity score matching model, they analysed the ease of issuing a green bond vis-à-vis other conventional bonds of the same nature. They found that it is easier to issue green bonds than non-green bonds, which raises the question of why Indian companies are not actively participating in green-project financing, or why there are fewer Indian green-bond issuers. Wiśniewski and Zieliński (2019) stated that the green-bonds market has grown over the last 10 years but still has not developed completely, which means that there is scope for more growth. They also mentioned that green-bond issues have improved both the investor's and issuer's image as being socially responsible. Further, they found that potential investors may be easily attracted through issuing green bonds instead of other financial instruments. Agliardi and Agliardi (2019) presented a structural model for pricing green bonds and coined the term 'greenium', which is the difference of yields between conventional and green bonds that are similar in characteristics. They also mentioned some variables that play an important role in the pricing of green bonds; for instance, issuing green bonds increases the credit rating of the issuer, which in return leads to decreasing the green bonds' cost of capital. This emphasizes that green bonds can help increase the value of a firm. However, problems may arise when a company needs funds to finance non-green projects, as proceeds from green bonds cannot be diverted. Nanayakkara and Colombage (2019) used the panel data regression model and option-adjusted spread to find out whether investors were paying a higher premium for green bonds. They found that investors were paying 63 basis points extra for green bonds as compared to conventional bonds. Green bonds not only motivate the issuers to raise funds but also help the investors diversify their portfolio for better returns. Baulkaran (2019) studied how the stock return of the issuer was affected by the issue of green bonds. The findings show that the cumulative abnormal return (CAR) was positive and significant. The researcher also stated that investors react negatively when green bonds are issued with a higher coupon rate. This study signifies the inverse relationship between bond price and interest rate. A higher coupon rate may decrease the value of a green bond, which shows that the coupon rate should be optimum, so that it does not affect the market price. Barua and Chiesa (2019) explored the factors affecting the issue size of green bonds. These factors were identified as characteristics of the bond, the issuer and the market. In terms of market, they explored that average issue size did not remain consistent over time, and high-grade bond average issue size was significantly lower than medium-grade bond. Thus, deciding the issue size of a bond can be a challenge for the issuer and requires proper analysis. According to Zerbib (2019), green bond yields are lower

than those of conventional bonds, and particularly which fall into the lower grade bond category. The point was made that this impact reduces the appeal of pro-environmental preferences to investors, resulting in low investor participation in the development of the green bond market. This means those investors who do not have much concern for the environment would avoid investing in lower-grade bonds.

From the literature, we observe how green bonds affect the performance of a company in terms of capital and equity costs, interest rate, yields, issue size, etc. However, there is scope for further research on how the performance of companies gets affected by their issue of green bonds in terms of stock market returns. This study explores the effect of green bonds on the stock performance of the Indian institutions that have issued them. It investigates whether or not issuing green bonds is beneficial to businesses in terms of stock valuation, as well as whether or not the returns increase afterwards.

In the next section, the data and methodology of the study are discussed, after which we present the results and discussions. Implications for the future and conclusions are then outlined in the final sections of the article.

Data and Methodology

Data

The analysis of this article is based on secondary data. The prices of individual stocks are extracted from NSE. Information regarding the dates of issue of green bonds has been taken from different articles in the newspapers such as *Business Standard*, *The Financial Express*, *The Hindu*, *BusinessLine*, *Livemint*, *Business Today* and *The Economic Times*, and the same has been verified from Bonds &

Climate Change Newsletter and the Centre for Monitoring Indian Economy (CMIE) Industry Outlook database. In our initial sample, we considered 20 companies that issued green bonds, from a The Energy and Resources Institute (TERI) analysis report. From this sample, 10 banking and financial companies were shortlisted for the study, as mostly the financial sector issues green bonds to finance renewable energy. Finally, six companies were selected for data analysis. The other four companies were excluded due to lack of historical data of stock prices, as their stock was not listed on the exchange index. Out of the six companies considered, four are banking companies—Yes Bank, IDBI Bank, Axis Bank, SBI—and two are financial-services companies—REC and PFC.

Methodology

The event study method has been applied in the study to empirically investigate corporate events and also stock market reaction (Werner, 2010). This method is based on the hypothesis of semi-strong market efficiency. According to this hypothesis, it is impossible to get excess returns regularly, as new publicly available information is immediately reflected in the price. This methodology has become popular in the research community as a reliable tool due to its basic outline. The event study method involves comparing the performance of the event before and after the significant event and also examining the variables like trading volumes and return variances (Khotari & Warner, 2006). Figure 2 shows the event study timeline, which has been divided into two parts: the estimation period and the observation period. In the observation period, day 0 is the event day, which can be the announcement day of a particular event, such as a merger, demonetization or lockdown announcement.

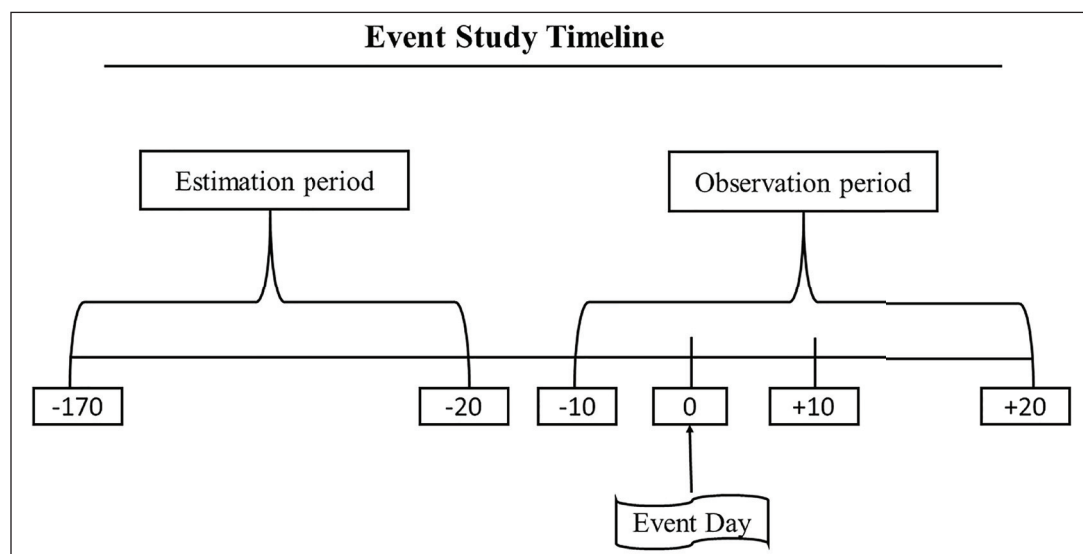


Figure 2. Event Study Timeline.

Source: The authors.

Using day 0 as a reference point, abnormal returns are calculated. If the abnormal returns are observed after the event, this means that the market is not capturing the information quickly, showing the possibility of making an abnormal profit. On the other hand, when abnormal returns are observed before the event, it is indicated that there might have been some insider information. The estimation period is used to calculate the stock's average return before the event to be used in the mean-adjusted stock returns model.

Before analysing the data, we calculated the mean return for each firm based on the daily returns from day -170 through -21 after taking a logarithmic return, where day 0 is the event day on which the green bond was issued. In Annexure 1, the descriptive summary is presented.

First, we calculated the abnormal return using the mean-adjusted returns model using the following formula given by Neuhierl et al. (2012):

$$AR_i = R_i - R_j,$$

where AR_i is the abnormal return on a given day, and R_i is the actual log return of the stock on a given day. The daily log return was calculated using the following formula:

$$R_i = \ln(P_i - [P_i - 1]),$$

where P_i and $(P_i - 1)$ represent the daily stock price at days t and $(t - 1)$, respectively. R_j is the average return of the stock, calculated taking the log return of the stock price.

After that, we calculated abnormal returns using the market-adjusted returns model. The market return was calculated from the stock price of NIFTY 50 according to the following formula:

$$AR_i = R_i - R_{M,t},$$

where $R_{M,t}$ is the market return on a given day and AR_i and R_i are the same as defined earlier.

Finally, we used a risk-adjusted returns model for robust results and to support the results obtained through the previous models, using the following formula:

$$AR_i = R_i - (\alpha + \beta R_{M,t}),$$

where α and β are estimates from a regression of the stock against the market index and AR_i and R_i are the same as defined earlier.

The CAR was calculated by adding the abnormal return from day -1 to day -10 and that from day 1 to day 10. For the analysis of data, we used the event study method. An event window of 21 days was taken for the study (from day

-10 to day 10, where day 0 is the date of green-bond issue). All the calculations were performed using R software.

Empirical Results and Discussions

Table 3 shows the CAR before and after the issue of green bonds. We observe that on the issue date it is negative for all companies except REC. Before the issue of green bonds, that is, from day -1 to day -10, the CAR is negative for all the samples taken for the study except Axis Bank, which shows a positive CAR for this period. After the green-bond issue, that is, from day 1 to day 10, the CAR is positive for all companies except Axis Bank and PFC, which show negative returns. The opposite outcomes for Axis Bank and PFC might be due to accessing of insider information before the issue of the green bonds. We used three methods for calculating the abnormal returns for robust results. All three, that is, the mean-adjusted returns model, market-adjusted returns model and risk-adjusted returns model, show the same outcomes after the event day. However, SBI is an exception, as the mean-adjusted returns model exhibits a negative return, though the market-adjusted returns model and risk-adjusted returns model show positive results.

Certainly, issuing green bonds increases investors' wealth in terms of capital gains by increasing the share price of the company. Yes Bank's performance was better than that of all the other institutions, with a return of 5.3% after the date of the issue of green bonds. We find that green bonds have a significant impact on the stock price returns.

Furthermore, at a 5% significance level, t -test is applied to examine if the mean difference in stock return before and after the issue of a green bond differs or not significantly. From Table 4, it is clear that the p -value is more than 0.05 for all companies. We reject the null hypothesis when the p -value is less than 0.05, which indicates the existence of mean difference in the sample. Here, we do not reject the null hypothesis, as a significant impact is indicated after the event for all samples. The results are consistent with Baulkaran's (2019) outcomes, with the stock market positively impacted after the issue of green bonds. Using the event study method, Hoa Nguyen et al. (2019) also found a significant positive effect on Vietnam's stock market due to the announcement

Table 3. Cumulative Abnormal Returns of the Banks and Financial Institutions (in %).

Bank/FI	MAR			MKAR			RAR		
	{-1, -10}	0	{+1, +10}	{-1, -10}	0	{+1, +10}	{-1, -10}	0	{+1, +10}
Yes Bank	-5.81	-2.61	5.30	-3.19	-2.39	6.22	-4.12	-2.51	4.54
IDBI Bank	-1.26	-0.92	4.33	0.84	-0.74	7.07	-0.25	-0.87	6.53
Axis Bank	9.88	-1.84	-2.18	4.18	-1.58	-2.43	2.34	-1.53	-2.70
REC	-10.55	1.03	4.58	-7.32	1.05	2.84	-7.69	0.84	-0.49
PFC	-2.14	-0.54	-6.78	-3.83	-0.48	-5.40	-2.85	-0.31	-3.46
SBI	-11.27	-0.85	-5.00	-8.58	-0.45	3.56	-6.19	-6.19	9.10

Source: The authors.

Note: FI—Financial institution; MAR—mean-adjusted returns model; MKAR—market-adjusted returns model; and RAR—risk-adjusted returns model.

Table 4. *t*-Test Results for the Banks and Financial Institutions.

Bank/FI	MAR			MKAR			RAR		
	CAR (I, 10)	<i>t</i> -Test	<i>p</i> -Value*	CAR (I, 10)	<i>t</i> -Test	<i>p</i> -Value*	CAR (I, 10)	<i>t</i> -Test	<i>p</i> -Value*
Yes Bank	5.30	−1.1070	0.2835	6.22	−1.1435	0.2680	4.54	−1.0985	0.2865
IDBI Bank	4.33	−0.4233	0.6777	7.07	−0.5039	0.6209	6.53	−0.5439	0.5933
Axis Bank	−2.18	1.9298	0.0758	−2.43	1.1098	0.2847	−2.70	0.8052	0.4326
REC	4.58	−1.9072	0.0728	2.84	−1.4197	0.1729	−0.49	−1.0429	0.3108
PFC	−6.78	0.8722	0.3946	−5.40	0.5419	0.5946	−3.46	0.3333	0.7428
SBI	−5.00	−0.6544	0.5216	3.56	−1.7360	0.1003	9.10	−2.2777	0.0355

Source: The authors.

Note: * Indicates significance at 5%.

of stock repurchase. Girish (2019) examined the effect of the goods and services tax on NIFTY 50 and found significant positive results: It generated abnormal returns during the 14 days before and after the event. Lodha et al. (2018) and Iyengar et al. (2017) took demonetization as an event in their study and showed that the banking, FMCG, financial-services and automobile sectors had generated positive returns.

Limitations and Implications for the Future

The samples considered in this study are few and limited to the banking and financial sectors. Future research involving different sectors of the economy can be conducted using models other than the ones employed in this study (MAR, MKAR, and RAR). The event study method has been used in this study. Other research techniques can be used for establishing the robustness of the results, like regression analysis. Comparison of different green-bond issuers and determination of their bond yields can be an exciting area of research.

Investors and companies should support sustainable development and improve the quality of the environment for the current and future generations. It may be achieved through the issuance of green bonds, which have a strong brand value associated with environmentally friendly projects. It will increase the wealth of companies and investors and simultaneously improve the quality of the environment too.

Challenges for Entities Working in India

Indian issuers face many challenges in issuing green bonds. Some significant ones are mentioned below (Kala & Garg, 2015).

- Hedging costs: The cost of hedging is very high, pushing issuers to find alternate sources of funding to reduce the capital costs.
- Credit ratings: Credit rating is the most critical factor for investors, especially for the risk-averse ones. Credit rating from such agencies as International Finance Corporation (IFC) and Agence Française de

Développement (AFD) comes at very high costs. This requires a proper cost–benefit analysis by the entities.

- Regulations: There are specific restrictions on using proceeds from external commercial borrowings. For instance, only 25% of the proceeds can be used for existing projects; the remaining 75% should be reserved for the new projects, which may hamper existing projects' operation.

Conclusion

India has growth potential in the green-bond market, as the economy of India is growing. Investors are also more inclined towards green bonds, which have been oversubscribed when issued. Most of the green-bond proceeds issued by Indian institutions are utilized in renewable-energy generation (Agarwal & Singh, 2018). Indian financial institutions and corporations may issue green bonds to finance other green projects, such as green buildings, low-carbon transportation, clean water, waste management and sustainable land use.

We observe that green bonds issued by Indian banks and financial institutions significantly affect the stock returns of these institutions. This increases the wealth of the shareholders and the value of the firms. This study may motivate other corporations to issue green bonds to show their concern for the environment and climate change. Investors may also opt for such investment, which will help diversify their portfolio and provide them more options for safe returns.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

Funding

The authors received no financial support for the research, authorship and/or publication of this article.

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> summary(data.frame(`Nifty SBI`, SBI, `Nifty PFC`, PFC, `Nifty REC`, REC, `Nifty AXIS`, `AXIS BANK`, `Nifty I
DBI`, IDBI, `Nifty Yes bank`, `Yes Bank`))
  Nifty.SBI      SBI      Nifty.PFC      PFC      Nifty.REC
Min.   :-0.023539 Min.   :-5.211e-02 Min.   :-0.0156827 Min.   :-0.0843441 Min.   :-0.0272814
1st Qu.: -0.003693 1st Qu.: -1.392e-02 1st Qu.: -0.0021208 1st Qu.: -0.0141160 1st Qu.: -0.0027121
Median :  0.000637 Median : -1.581e-03 Median :  0.0006935 Median :  0.0019005 Median :  0.0006935
Mean   :  0.000593 Mean   : -3.946e-05 Mean   :  0.0008050 Mean   : -0.0002568 Mean   :  0.0005953
3rd Qu.:  0.005267 3rd Qu.:  1.235e-02 3rd Qu.:  0.0046950 3rd Qu.:  0.0128933 3rd Qu.:  0.0046433
Max.   :  0.019239 Max.   :  5.667e-02 Max.   :  0.0169190 Max.   :  0.0721375 Max.   :  0.0185082

      REC      Nifty.AXIS      AXIS.BANK      Nifty.IDBI      IDBI
Min.   :-0.062434 Min.   :-3.373e-02 Min.   :-0.0774748 Min.   :-0.0609726 Min.   :-0.1613799
1st Qu.: -0.011590 1st Qu.: -5.681e-03 1st Qu.: -0.0125828 1st Qu.: -0.0056449 1st Qu.: -0.0176683
Median :  0.002330 Median :  1.541e-04 Median : -0.0004301 Median : -0.0007436 Median :  0.0000000
Mean   :  0.002029 Mean   :  4.711e-05 Mean   :  0.0002049 Mean   : -0.0007015 Mean   :  0.0008422
3rd Qu.:  0.015610 3rd Qu.:  6.087e-03 3rd Qu.:  0.0109517 3rd Qu.:  0.0060047 3rd Qu.:  0.0172304
Max.   :  0.121455 Max.   :  3.312e-02 Max.   :  0.0656805 Max.   :  0.0209588 Max.   :  0.1527139

  Nifty.Yes.bank      Yes.Bank
Min.   :-0.030422 Min.   :-0.053124
1st Qu.: -0.002855 1st Qu.: -0.010864
Median :  0.001791 Median :  0.001955
Mean   :  0.001131 Mean   :  0.002578
3rd Qu.:  0.006118 3rd Qu.:  0.015461
Max.   :  0.025831 Max.   :  0.053080

```

Annexure I. Descriptive Summary of Data Using R.

Source: Authors' own calculation.

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